

ASSIGNMENT BRIEF

HTU Course No: 40201220	HTU Course Name: Software Development Lifecycles
BTEC Unit No: K/618/7408	BTEC UNIT Name: Unit 7 Software Development Lifecycles

Version: 3



Student Name/ID Number/Section	
HTU Course Number and Title	40201220 Software Development Lifecycles
BTEC Unit Number and Title	K/618/7408 Unit 7 Software Development Lifecycles
Academic Year	2023-2024 Fall
Assignment Author	Salem Alemaishat
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Assignment Title	Building a Software Utilizing a Software Development Lifecycle & Model
Assignment Ref No	1
Issue Date	25/11/2023
Formative Assessment dates	From 24/12/2023 to 11/01/2024
Submission Date	27/01/2024
IV Name & Date	Orwa Aladaileh 24/11/2023
Submission Format	
Submission for this assignment is expected to be an <u>individual written report</u>. This report should be: In the form of a soft copy (.docx) submitted to HTU Elearning, Signed Declaration form (Included) Written in a formal business style (headings, content page, paragraphs, subsections, and illustrations as appropriate, single spacing & font size 12) Supported with research and referenced using the Harvard referencing system. An oral discussion with your assessors illustrating your assignment and answering questions	
Unit Learning Outcomes	
LO1 Describe different software development lifecycles LO2 Explain the importance of a feasibility study LO3 Undertake a software development lifecycle LO4 Discuss the suitability of software behavioural design techniques	
Assignment Brief and Guidance	
Scenario: You are an IT professional employed as a “software engineer” at “App Tech”, a company specializing in application development of all types. You have been tasked with building a fitness app “FitPro”. Profit will help users improve their fitness, offer personalized fitness plans, and nutrition guidance. FitPro includes tailored exercise and meal plans, a vibrant user community, progress tracking, and additional features like live workouts and expert articles. You will be required to provide the following: Part 1 Report	

Your report should **describe** two iterative and two sequential software development lifecycle models. Furthermore, you need to **explain** how risk is being managed in the software development lifecycle and **discuss** using example(s) a particular lifecycle model for a specific development environment and select the fit model for FitPro and justify your answer. You will also need to assess the benefits of using the Waterfall model in a large software development project (related to real-life projects).

Your report should **describe** how technical solutions can be compared (compare two). Next, **explain** the purpose of the feasibility report, discuss its components and assess the impact of different feasibility criteria on a software investigation.

Using examples, discuss the suitability of software behavioral design techniques and analyze a range of software behavioral tools and techniques (two tools and two techniques). Your report should differentiate between a finite state machine (FSM) and an extended FSM by providing an application of use for both. Finally, justify how data-driven software can improve the reliability and effectiveness of software.

Part 2 Software

After collecting user requirements, investigate whether the software meets a business need. You should **use** appropriate software analysis tools/techniques to carry out a software investigation and create supporting documentation. You will also need to analyze how software requirements can be traced throughout the software lifecycle and discuss two approaches to improving software quality. Finally, you need to evaluate the process of **undertaking** a systems investigation with regard to its effectiveness in improving software quality.

In summary, the student must submit:

- A final report based on the requirements in Part 1 & 2:
- The software part includes:
 - o Feasibility study of the proposed idea.
 - o Software Requirement Specification based on IEEE 830 (or any similar standard)
 - o A requirements traceability matrix (RTM)
 - o Software Design Specification (complimentary and detailed design)

Learning Outcomes and Assessment Criteria			
Learning Outcome	Pass	Merit	Distinction
LO1 Describe different software development lifecycles	P1 Describe two iterative and two sequential software lifecycle models. P2 Explain how risk is managed in software lifecycle models.	M1 Discuss using an example, why a particular lifecycle model is selected for a development environment.	D1 Assess the merits of applying the Waterfall lifecycle model to a large software development project.
LO2 Explain the importance of a feasibility study	P3 Explain the purpose of a feasibility report. P4 Describe how technical solutions can be compared.	M2 Discuss the components of a feasibility report.	D2 Assess the impact of different feasibility criteria on a software investigation.
LO3 Undertake a software development lifecycle	P5 Undertake a software investigation to meet a business need. P6 Use appropriate software analysis tools/techniques to carry out a software investigation and create supporting documentation.	M3 Analyse how software requirements can be traced throughout the software lifecycle. M4 Discuss two approaches to improving software quality.	D3 Evaluate the process of undertaking a systems investigation with regard to its effectiveness in improving software quality.
LO4 Discuss the suitability of software behavioural design techniques	P7 Discuss, using examples, the suitability of software behavioural design techniques.	M5 Analyse a range of software behavioural tools and techniques. M6 Differentiate between a finite state machine (FSM) and an extended FSM, providing an application of use for both.	D4 Present justifications of how data-driven software can improve the reliability and effectiveness of software.

STUDENT ASSESSMENT SUBMISSION AND DECLARATION

When submitting evidence for assessment, each student must sign a declaration confirming that the work is their own

Student name:	Assessor name:	
Issue date: 25/11/2023	Submission date: 27/01/2024	Submitted on:
Programme: Computing		
HTU Course Name: Software Development Lifecycles HTU Course Code: 40201220	BTEC Course Title: Unit 7 Software Development Lifecycles BTEC Course Code: K/618/7408	
Assignment number and title: 1, Building a Software Utilizing a Software Development Lifecycle & Model		

Plagiarism:

Plagiarism is a particular form of cheating. Plagiarism must be avoided at all costs and students who break the rules, however innocently, may be penalised. It is your responsibility to ensure that you understand **correct referencing practices**. As a university level student, you are expected to use appropriate references throughout and keep carefully detailed notes of all your sources of materials for material you have used in your work, including any material downloaded from the Internet. Please consult the relevant unit lecturer or your course tutor if you need any further advice.

I certify that the assignment submission is entirely my own work and I fully understand the consequences of plagiarism. I understand that making a false declaration is a form of malpractice.	
Student Name:	Student Signature:
	Date: